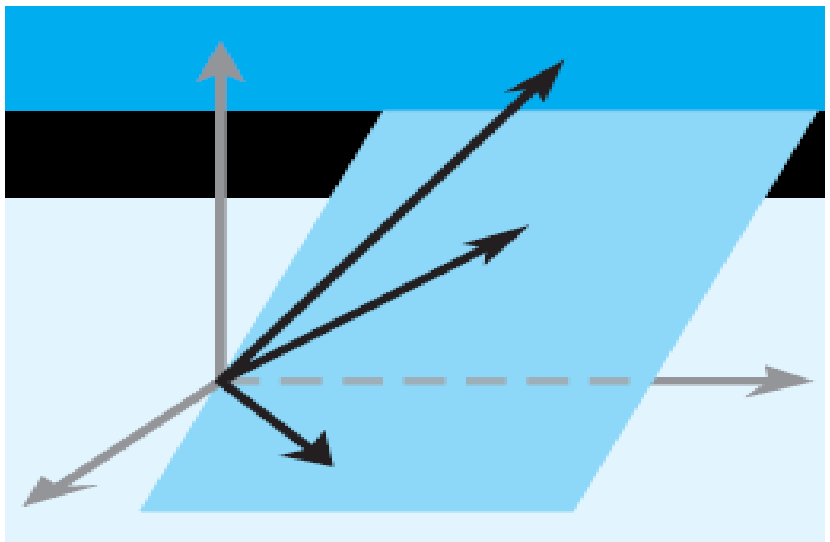
Elementary Linear Algebra

Chapter 7



Howard Anton

Chapter 7
Diagonalization & Quadratic Forms

Section 7.3 Quadratic Forms

$$a_1x_1^2 + a_2x_2^2 + 2a_3x_1x_2$$

$$a_1x_1^2 + a_2x_2^2 + a_3x_3^2 + 2a_4x_1x_2 + 2a_5x_1x_3 + 2a_6x_2x_3$$

Section 7.3 Quadratic Forms

EXAMPLE 1 Expressing Quadratic Forms in Matrix Notation

In each part, express the quadratic form in the matrix notation $\mathbf{x}^T A\mathbf{x}$, where A is symmetric.

(a) $2x^2 + 6xy - 5y^2$

(b) $x_1^2 + 7x_2^2 - 3x_3^2 + 4x_1x_2 - 2x_1x_3 + 8x_2x_2$

Solution

The diagonal entries of A are the coefficients of the squared terms, and the off-diagonal entries are half the coefficients of the cross product terms, so

$$2x^2 + 6xy - 5y^2 = \begin{bmatrix} x & y \end{bmatrix} \boxed{\quad\quad} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x_1^2 + 7x_2^2 - 3x_3^2 + 4x_1x_2 - 2x_1x_3 + 8x_2x_3 = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \boxed{\quad\quad\quad} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Section 7.3 Quadratic Forms

There are three important kinds of problems that occur in applications of quadratic forms:

Problem 1 If $\mathbf{x}^T A\mathbf{x}$ is a quadratic form on R^2 or R^3, what kind of curve or surface is represented by the equation $\mathbf{x}^T A\mathbf{x} = k$?

Problem 2 If $\mathbf{x}^T A\mathbf{x}$ is a quadratic form on R^n, what conditions must A satisfy for $\mathbf{x}^T A\mathbf{x}$ to have positive values for $\mathbf{x} \neq \mathbf{0}$?

Problem 3 If $\mathbf{x}^T A\mathbf{x}$ is a quadratic form on R^n, what are its maximum and minimum values if $\mathbf{x}$ is constrained to satisfy $\|\mathbf{x}\| = 1$?

Section 7.3 Quadratic Forms

A is symmetric matrix, by Theorem 7.2.1, A is orthogonal diagonalizable $P^T A P = D$

$$\mathrm{x}^T A \mathrm{x}$$

$$= \mathrm{x}^T PDP^T \mathrm{x}$$

$$= \mathrm{y}^T D \mathrm{y}$$

$$\boxed{\begin{array}{l} P^T \mathrm{x} = \mathrm{y} \\ \mathrm{x} = P\mathrm{y} \end{array}}$$

$$\mathrm{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\mathrm{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

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A **is symmetric matrix,** **by Theorem 7.2.1**, ***A*** **is orthogonal diagonalizable** $P^T A P = D$

$$\begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \mathrm{y}^T D \, \mathrm{y}$$

$$= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

Section 7.3 Quadratic Forms

THEOREM 7.3.1 The Principal Axes Theorem

If A is a symmetric $n \times n$ matrix, then there is an orthogonal change of variable that transforms the quadratic form $\mathbf{x}^T A\mathbf{x}$ into a quadratic form $\mathbf{y}^T D\mathbf{y}$ with no cross product terms. Specifically, if P orthogonally diagonalizes A, then making the change of variable $\mathbf{x} = P\mathbf{y}$ in the quadratic form $\mathbf{x}^T A\mathbf{x}$ yields the quadratic form

$P^T \mathrm{x} = \mathrm{y}$

$$\mathbf{x}^T A\mathbf{x} = \mathbf{y}^T D\mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

in which $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A corresponding to the eigenvectors that form the successive columns of P.

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EXAMPLE 2 **An Illustration of the Principal Axes Theorem**

Find an orthogonal change of variable that eliminates the cross product terms in the quadratic form $Q = x_1^2 - x_3^2 - 4x_1x_2 + 4x_2x_3$, and express Q in terms of the new variables.

Solution

The quadratic form can be expressed in matrix notation as

$$Q = \mathbf{x}^T A\mathbf{x} = [x_1 \;\; x_2 \;\; x_3] \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The characteristic equation of the matrix A is

$$\begin{vmatrix} \lambda - 1 & 2 & 0 \\ 2 & \lambda & -2 \\ 0 & -2 & \lambda + 1 \end{vmatrix} = \lambda^3 - 9\lambda = \lambda(\lambda + 3)(\lambda - 3) = 0$$

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so the eigenvalues are $\lambda = 0,\ -3,\ 3$. We leave it for you to show that orthonormal bases for the three eigenspaces are

$$\lambda = 0: \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}, \quad \lambda = -3: \begin{bmatrix} -\frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{bmatrix}, \quad \lambda = 3: \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

Thus, a substitution $\mathbf{x} = P\mathbf{y}$ that eliminates the cross product terms is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \boxed{\begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

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This produces the new quadratic form

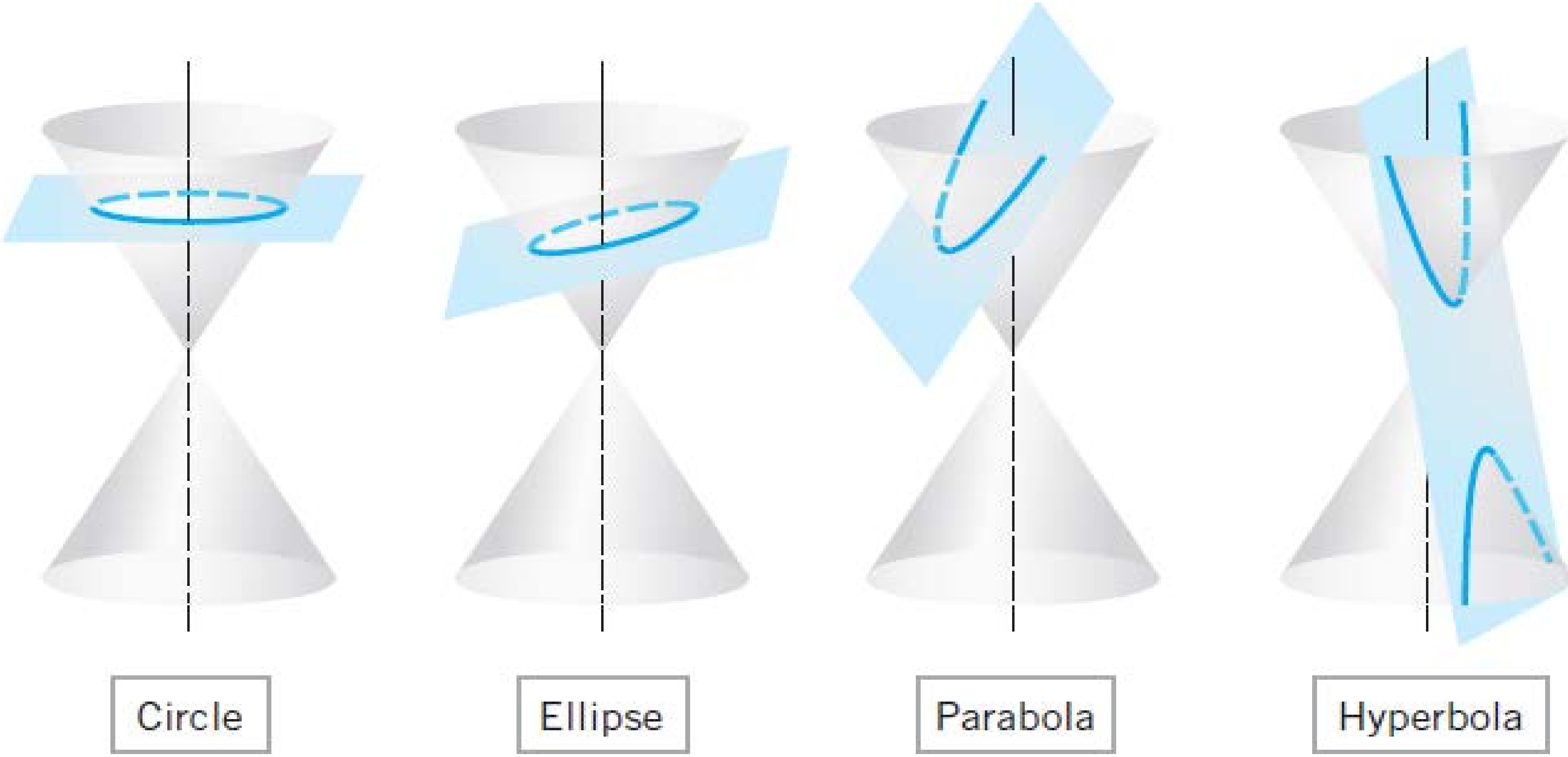
$$Q = \mathbf{y}^T\left(P^T A P\right)\mathbf{y} = [y_1 \quad y_2 \quad y_3]\begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = -3y_2^2 + 3y_3^2$$

in which there are no cross product terms.

Section 7.3 Quadratic Forms

Conic Sections

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0$$



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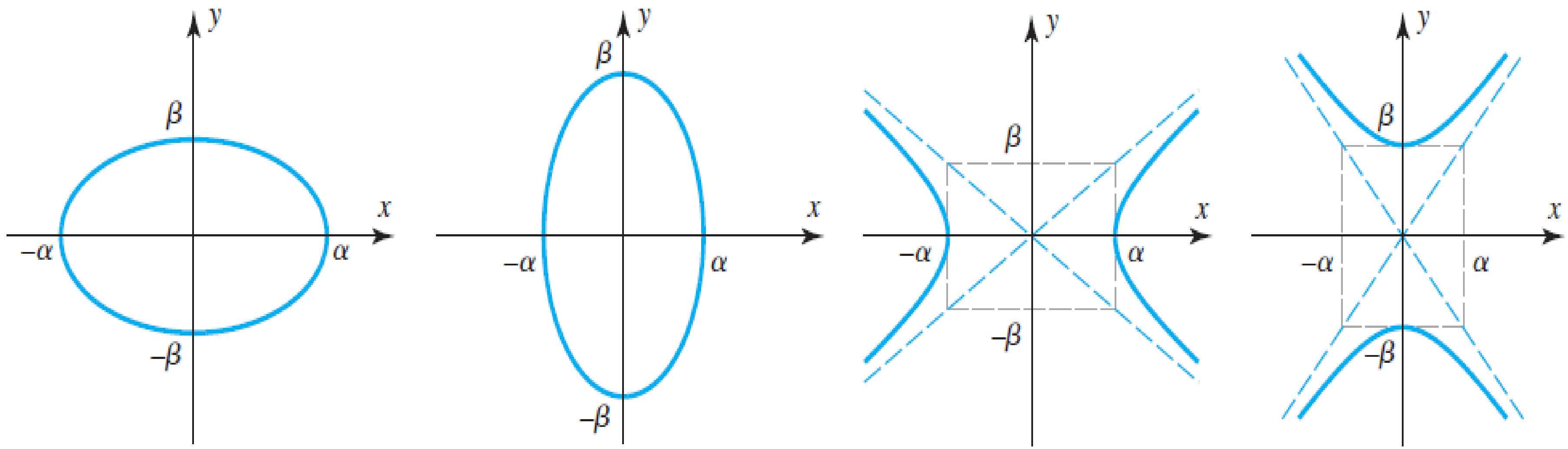
Central conics $ax^2 + 2bxy + cy^2 + f = 0$

Central conics in standard position

$$ax^2 + cy^2 + f = 0$$

Table 1



$\dfrac{x^2}{\alpha^2} + \dfrac{y^2}{\beta^2} = 1$ $(\alpha \geq \beta > 0)$	$\dfrac{x^2}{\alpha^2} + \dfrac{y^2}{\beta^2} = 1$ $(\beta \geq \alpha > 0)$	$\dfrac{x^2}{\alpha^2} - \dfrac{y^2}{\beta^2} = 1$ $(\alpha > 0, \beta > 0)$	$\dfrac{y^2}{\beta^2} - \dfrac{x^2}{\alpha^2} = 1$ $(\alpha > 0, \beta > 0)$

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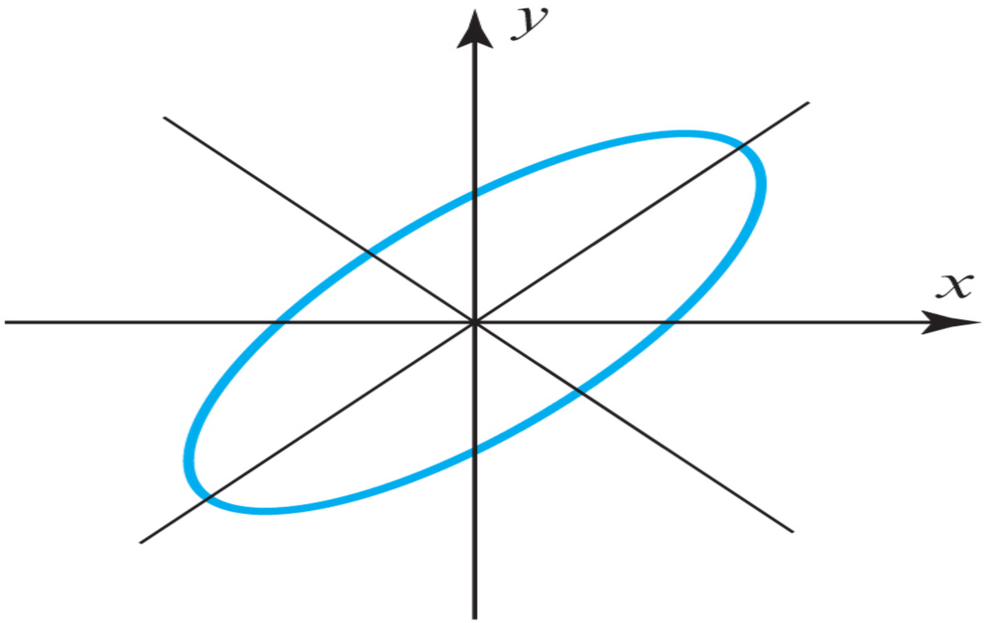
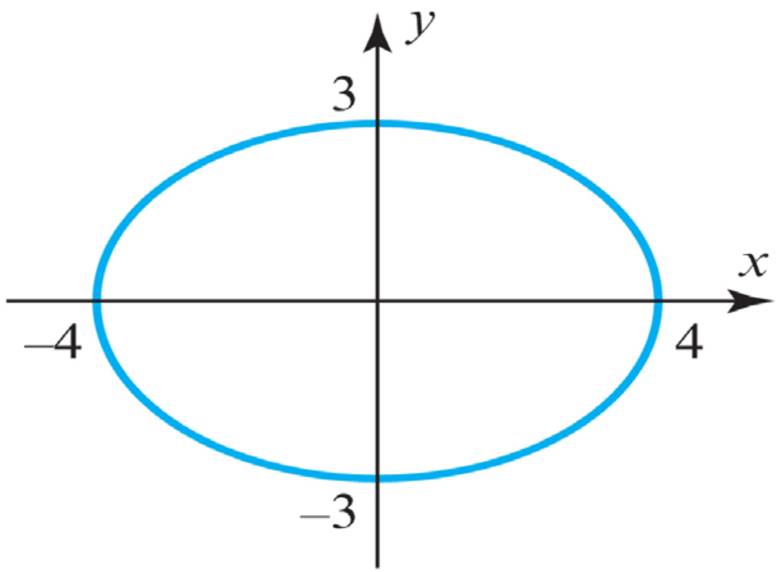
$$ax^2 + 2bxy + cy^2 + f = 0$$

$$ax^2 + cy^2 + f = 0$$



A central conic rotated out of standard position



$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

Central conics in standard position

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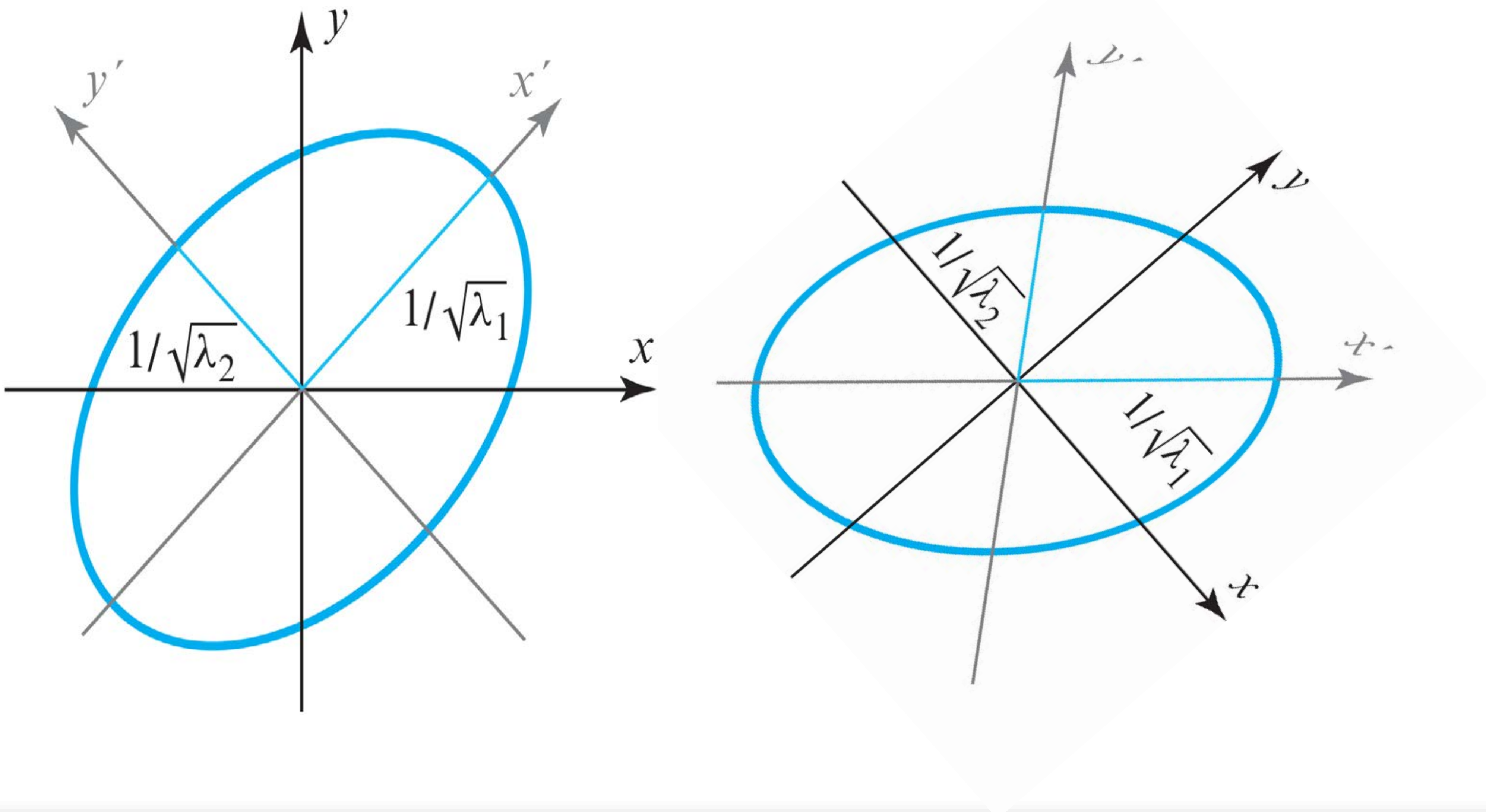

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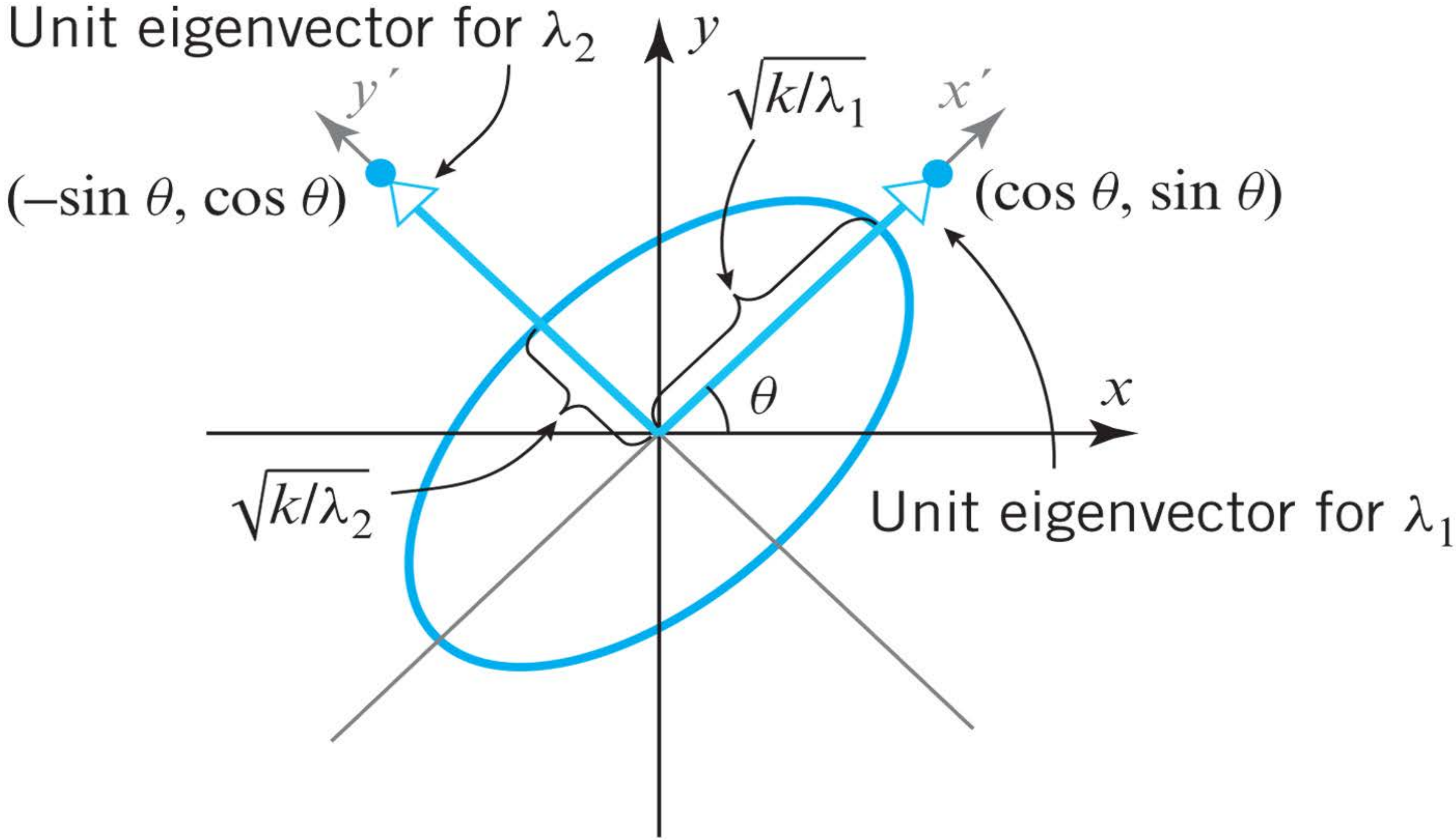

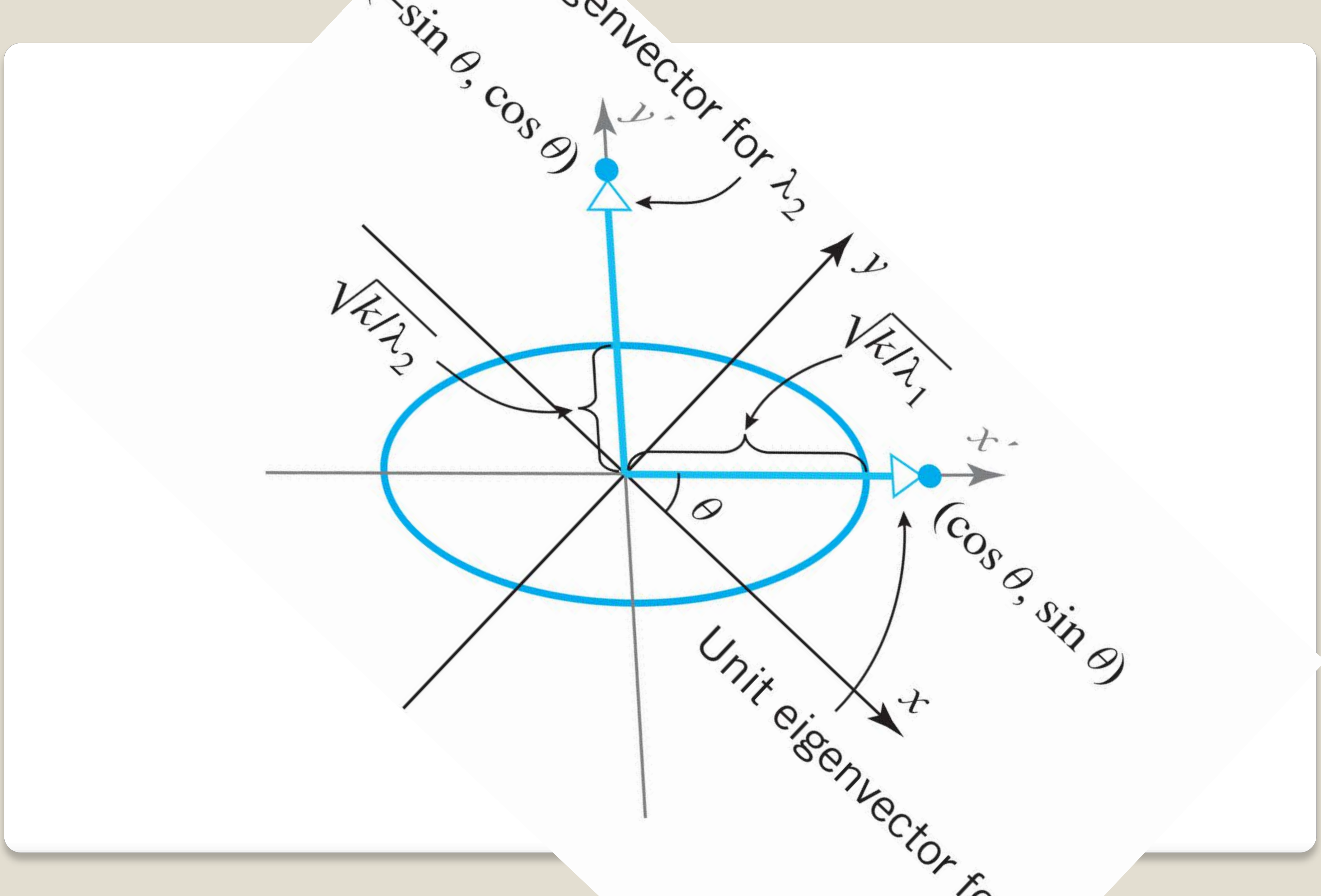

$\sin\theta, \cos\theta)$
for λ_2
y'
y
$\sqrt{k/\lambda_2}$
$\sqrt{k/\lambda_1}$
x'
θ
$(\cos\theta, \sin\theta)$
x
Unit eigenvector

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EXAMPLE 3 Identifying a Conic by Eliminating the Cross Product Term

(a) Identify the conic whose equation is $5x^2 - 4xy + 8y^2 - 36 = 0$ by rotating the xy-axes to put the conic in standard position.

(b) Find the angle θ through which you rotated the xy-axes in part (a).

Solution (a) The given equation can be written in the matrix form

$$\mathbf{x}^T A\mathbf{x} = 36$$

where

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 8 \end{bmatrix}$$

The characteristic polynomial of A is

$$\begin{vmatrix} \lambda - 5 & 2 \\ 2 & \lambda - 8 \end{vmatrix} = (\lambda - 4)(\lambda - 9)$$

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so the eigenvalues are $\lambda = 4$ and $\lambda = 9$. We leave it for you to show that orthonormal bases for the eigenspaces are

$$\lambda = 4: \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}, \quad \lambda = 9: \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

Thus, A is orthogonally diagonalized by

$$P = \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \tag{18}$$

Moreover, it happens by chance that $\det(P) = 1$, so we are assured that the substitution $\mathbf{x} = P\mathbf{x}'$ performs a rotation of axes. It follows from 16 that the equation of the conic in the $x'y'$-coordinate system is

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$$\begin{bmatrix} x' & y' \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = 36$$

which we can write as

$$4x'^2 + 9y'^2 = 36 \quad \text{or} \quad \frac{x'^2}{9} + \frac{y'^2}{4} = 1$$

We can now see from Table 1 that the conic is an ellipse whose axis has length $2\alpha = 6$ in the x'-direction and length $2\beta = 4$ in the y'-direction.

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Solution (b) It follows from 15 that

$$P = \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

P has the effect of rotating the xy-axes of a rectangular coordinate system through an angle θ.

which implies that

$$\cos\theta = \frac{2}{\sqrt{5}}, \quad \sin\theta = \frac{1}{\sqrt{5}}, \quad \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{2}$$

Thus, $\theta = \tan^{-1}\frac{1}{2} \approx 26.6°$ (Figure 7.3.5).

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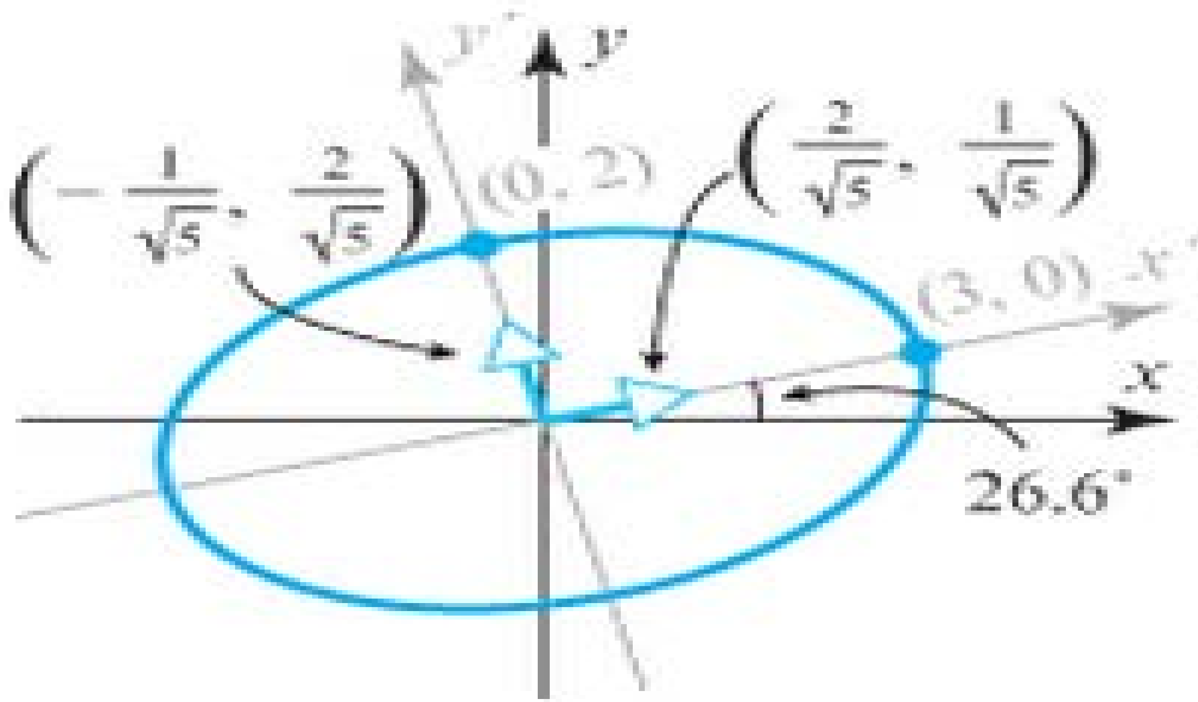

► Figure 7.3.5

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DEFINITION 1

A quadratic form $\mathbf{x}^T A\mathbf{x}$ is said to be

positive definite if $\mathbf{x}^T A\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$;

negative definite if $\mathbf{x}^T A\mathbf{x} < 0$ for $\mathbf{x} \neq \mathbf{0}$;

indefinite if $\mathbf{x}^T A\mathbf{x}$ has both positive and negative values.

DEFINITION 1

if $\mathbf{x}^T A\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$; **matrix A is positive definite**

if $\mathbf{x}^T A\mathbf{x} < 0$ for $\mathbf{x} \neq \mathbf{0}$; **matrix A is negative definite**

indefinite if $\mathbf{x}^T A\mathbf{x}$ has both positive and negative values.

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THEOREM 7.3.2 *If A is a symmetric matrix, then:*

(a) A *is positive definite if and only if all eigenvalues of A are positive.*

(b) A *is negative definite if and only if all eigenvalues of A are negative.*

(c) A *is indefinite if and only if A has at least one positive eigenvalue and at least one negative eigenvalue.*

Proof **Page 412**

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If all eigenvalues of A are positive then A is positive definite.

Proof By definition of positive definite, we want to show $\mathbf{x}^T A\mathbf{x} > 0$

A is symmetric matrix then

$P^T AP = D$ A can be orthogonal diagonalizable

$$\mathbf{x}^T A\mathbf{x}$$

$$= \mathbf{x}^T PDP^T \mathbf{x}$$

Because A is symmetric $A = PDP^T$

$$= \mathbf{y}^T D\mathbf{y}$$

by Principal axes theorem

$$= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

$$> 0$$

all eigenvalues of A are positive

A is positive definite

Section 7.3 Quadratic Forms

If A is positive definite then all eigenvalues of A are positive.

Proof **By definition of positive definite, we have** $x^T A x > 0$

By the definition of eigenvalues $Ax = \lambda x$

$$x^T A x$$

$$= x^T \lambda x$$ ***By definition of eigenvalues***

$$= \lambda x^T x$$

$$= \lambda \|x\|^2 > 0$$ ***A* is positive definite**

$\Rightarrow$ $\lambda > 0$

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EXAMPLE 4 Positive Definite Quadratic Forms

It is not usually possible to tell from the signs of the entries in a symmetric matrix A whether that matrix is positive definite, negative definite, or indefinite. For example, the entries of the matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

are nonnegative, but the matrix is indefinite since its eigenvalues are $\lambda = 1, 4, -2$ (verify). To see this another way, let us write out the quadratic form as

$$\mathbf{x}^T A\mathbf{x} = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 3x_1^2 + 2x_1x_2 + 2x_1x_3 + 4x_2x_3$$

We can now see, for example, that

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$$\mathbf{x}^T A\mathbf{x} = 4 \quad \text{for} \quad x_1 = 0, \quad x_2 = 1, \quad x_3 = 1$$

and

$$\mathbf{x}^T A\mathbf{x} = -4 \quad \text{for} \quad x_1 = 0, \quad x_2 = 1, \quad x_3 = -1$$

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THEOREM 7.3.3

If A is a symmetric 2×2 matrix, then:

(a) $\mathbf{x}^T A\mathbf{x} = 1$ *represents an ellipse if A is positive definite.*

(b) $\mathbf{x}^T A\mathbf{x} = 1$ *has no graph if A is negative definite.*

(c) $\mathbf{x}^T A\mathbf{x} = 1$ *represents a hyperbola if A is indefinite.*

THEOREM 7.3.4 *A symmetric matrix A is positive definite if and only if the determinant of every principal submatrix is positive.*

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EXAMPLE 5 Working with Principal Submatrices

The matrix

$$A = \begin{bmatrix} 2 & -1 & -3 \\ -1 & 2 & 4 \\ -3 & 4 & 9 \end{bmatrix}$$

is positive definite since the determinants

$$|2| = 2, \quad \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3, \quad \begin{vmatrix} 2 & -1 & -3 \\ -1 & 2 & 4 \\ -3 & 4 & 9 \end{vmatrix} = 1$$

are all positive. Thus, we are guaranteed that all eigenvalues of A are positive and $\mathbf{x}^T A\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$.

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Homework

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